Instrumental Variables

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Outline

- Why do we need IV?
- What is good IV?
- How to find good IV?
- How to estimate a model with IV?
- How to test whether an IV is good or not?

We will use the following paper to illustrate the above points

The Colonial Origins of Comparative Development: An Empirical Investigation
Author(s): Daron Acemoglu, Simon Johnson, James A. Robinson
Source: *The American Economic Review*, Vol. 91, No. 5 (Dec., 2001), pp. 1369-1401

Research Question

- What are the fundamental causes of the large differences in income per capita across countries?
- Better institutions: secure property rights, less distortionary policies -> more investment in physical and human capital

Do Institutions Matter for Income?

TABLE 2—OLS REGRESSIONS

	Whole world (1)	Base sample (2)	Whole world (3)	Whole world (4)	Base sample (5)	Base sample (6)	Whole world (7)	Base sample (8)
	Dependent variable is log GDP per capita in 1995						Dependent variable is log output per worker in 1988	
Average protection against expropriation risk, 1985–1995	0.54 (0.04)	0.52 (0.06)	0.47 (0.06)	0.43 (0.05)	0.47 (0.06)	0.41 (0.06)	0.45 (0.04)	0.46 (0.06)
Latitude			0.89 (0.49)	0.37 (0.51)	1.60 (0.70)	0.92 (0.63)		
Asia dummy				−0.62 (0.19)		−0.60 (0.23)		
Africa dummy				−1.00 (0.15)		−0.90 (0.17)		
"Other" continent dummy				−0.25 (0.20)		−0.04 (0.32)		
R^2	0.62	0.54	0.63	0.73	0.56	0.69	0.55	0.49
Number of observations	110	64	110	110	64	64	108	61

I. Why do we need IV?

Motivation

Three important threats to internal validity:

- Omitted variables (Unobserved, so cannot be included in the regression) – If not constant over time or states, panel data do not help
- Measurement error (X is measured with error)
- Simultaneity (X causes Y, Y causes X)

All of them imply that $E[u|X] \neq 0$, and OLS estimator is biased and inconsistent.

Instrumental variables (IV) regression can eliminate bias from these three sources.

Motivation

- Reverse causality: rich economies choose better institutions
- Omitted variable: economies that are different for a variety of reasons will differ both in their institutions and in their income per capita
- Measurement error: any measure only measures part of the institution, historical institution also matters, etc.

We need a source of exogenous variation in institutions!

II. What is a good IV?

What Is IV?

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

- Given that X is correlated with u, we can break X into two parts: a part that might be correlated with u, and a part that is not. By isolating the part that is not correlated with u, it is possible to estimate β_1
- This can be done by an instrumental variable (IV), denoted as Z, which is uncorrelated with u
- The IV Z detects movements in X that are uncorrelated with u, and thus estimate β_1

Two Conditions: Validity of IV

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

For an instrumental variable (an "instrument") Z to be valid, it must satisfy two conditions:

1. Relevance: $corr(Z, X) \neq 0$
2. Exogeneity: $corr(Z, u) = 0$

Alternatively, you may explain: Z affects Y only through X.

- IV is not simply a proxy for the causal variable of interest
- IV is not an omitted variable

Suppose for now that you have such a Z (we'll discuss how to find instrumental variables later). How can you use Z to estimate β_1?

III. How to find a good IV?

How to find a good IV?

- IV is not omitted variable
- IV is an “exogenous” shock
- Sometimes, policy shocks, natural shocks or historical shocks are exogenous enough
- There is no definite rule – use your judgement!

How to Find IV?

- Levitt (1997): effect of police on crime
  - Concern: high-crime areas require more police
  - IV: Mayor electoral cycle - affect the number of police without affecting crime
- Angrist and Krueger (1991): effect of schooling on future income
  - Concern: unobserved ability affects both schooling and income
  - IV: quarter of birth - affect years of schooling without affecting ability
- Angrist and Evans (1998): impact of childbirth on women labor participation
  - Concern: women with low earning potential have more kids and do not work
  - IV: gender mix of the first two kids - affect the number of children without affecting earning potential

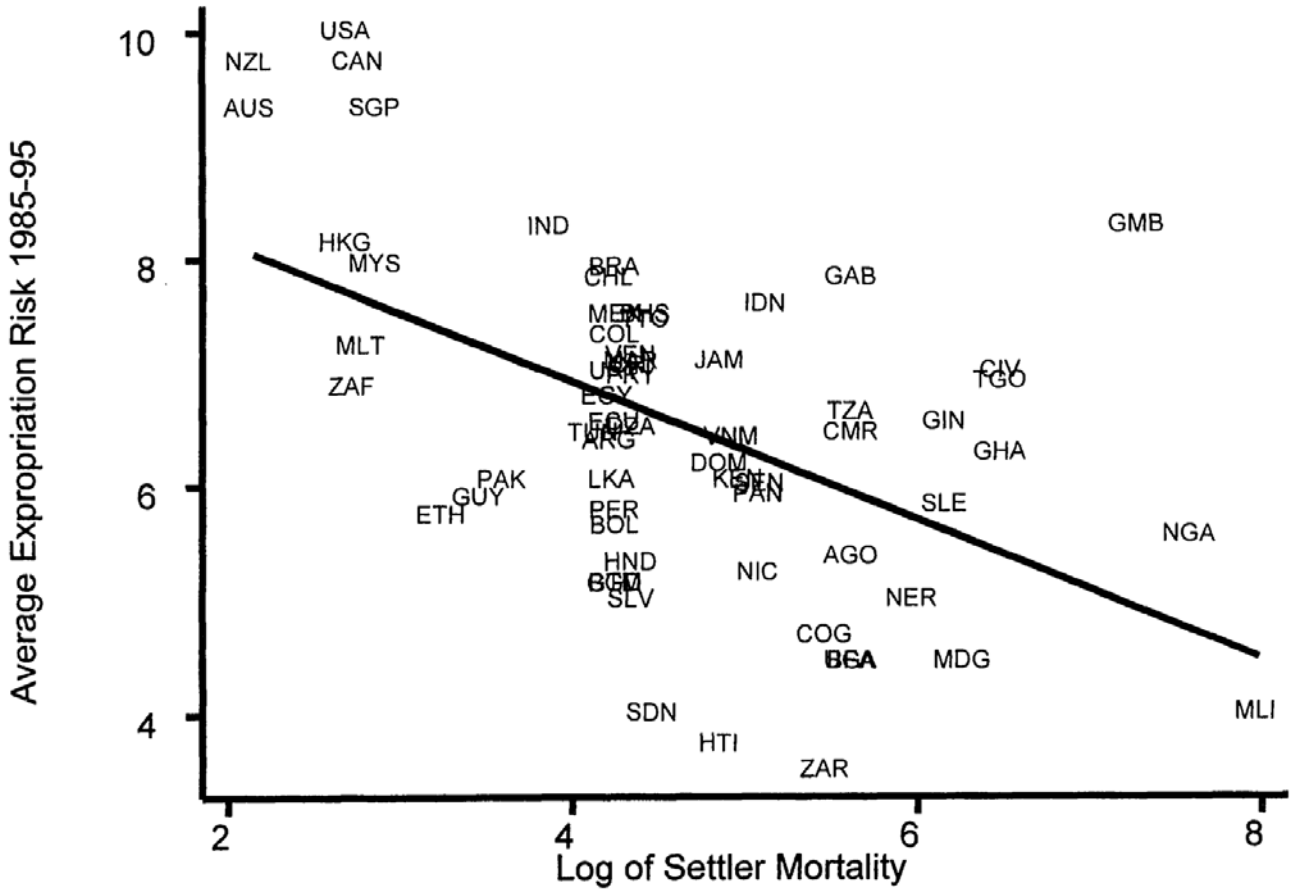
Early mortality rate as IV

(potential) settler mortality $\Rightarrow$ settlements

$\Rightarrow$ early institutions $\Rightarrow$ current institutions

$\Rightarrow$ current performance.

Relevance


Average Expropriation Risk 1985-95
10
8
6
4
2
4
6
8
Log of Settler Mortality
USA
NZL
CAN
AUS
SGP
HKG
MYS
IND
GMB
GAB
IDN
MLT
ZAF
JAM
CIV
TGO
TZA
CMR
GIN
GHA
PAK
LKA
GUY
ETH
SLE
NGA
AGO
HND
NIC
SLV
NER
COG
MDG
SDN
MLI
HTI
ZAR

Exogeneity

- Why not through health conditions?

> Most Black Africans and their descendants respond to yellow fever infection with mild to moderate symptoms such as headache, fever, nausea, and vomiting, and then recover in a few days. This outcome reflects the long relationship between the virus and its indigenous hosts, who through generations of exposure to the virus have evolved resistance.

In contrast, fatality rates among nonimmune adults, such as Europeans, could be as high as 90 percent.

IV. How to estimate a model with IV?

Two Stage Least Squares (2SLS)

Alternative explanation: (only) a little algebra

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

- From the exogeneity of Z ($cov(Z, u) = 0$), we have

$$\begin{aligned} 0 = cov(Z, u) &= cov(Z, Y - \beta_0 - \beta_1 X) \\ &= cov(Z, Y) - \beta_1 cov(Z, X) \end{aligned}$$

  - Exogeneity condition is the information you have

- From the relevance of Z ($cov(Z, X) \neq 0$), we have

$$\beta_1 = \frac{cov(Z, Y)}{cov(Z, X)}$$

  - Relevance condition ensures that we do not divide by zero

Two Stage Least Squares (2SLS)

$$\beta_1 = \frac{cov(Z, Y)}{cov(Z, X)}$$

- β_1 is equal to the ratio of two covariances
- Thus, we can estimate β_1 by replacing these population covariances with sample covariances – this is the method of moment estimator
- The resulting estimator is identical to $\hat{\beta}_{1,IV}$, so the IV estimator can be explained as the method of moment estimator

Note: If $Z = X$, the IV estimator becomes the OLS estimator!

General IV Regression

The general IV regression model:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_k X_{ki} + \beta_{k+1} W_{1i} + \cdots + \beta_{k+r} W_{ri} + u_i \, .$$

- Y_i is the dependent variable
- $X_{1i}, \ldots, X_{ki}$ are the endogenous regressors (potentially correlated with u_i)
- $W_{1i}, \ldots, W_{ri}$ are the control variables or included exogenous regressors/varibles (uncorrelated with u_i)
- $\beta_0, \beta_1, \ldots, \beta_{k+r}$ are the unknown regression coefficients
- $Z_{1i}, \ldots, Z_{mi}$ are the m instrumental variables (the excluded exogenous variables) ($m \geq k$ (why this requirement?))

Two Stage Least Squares (2SLS)

1. First, isolate the part of X_j that is uncorrelated with u:regress X_j on $Z_1, \ldots, Z_m$ and $W_1, \ldots, W_r$ using OLS

$$X_{ji} = \pi_{j,0} + \pi_{j,1}Z_{1i} + \cdots + \pi_{j,m}Z_{mi} + \pi_{j,m+1}W_{1i} + \cdots + \pi_{j,m+r}W_{ri} + v_{ji} \text{ for } j = 1, \ldots, k \quad (3)$$

   - Because $Z_{1i}, \ldots, Z_{mi}$ and $W_{1i}, \ldots, W_{ri}$ are uncorrelated with u_i, $\pi_{j,0} + \pi_{j,1}Z_{1i} + \cdots + \pi_{j,m}Z_{mi} + \pi_{j,m+1}W_{1i} + \cdots + \pi_{j,m+r}W_{ri}$ is uncorrelated with u_i. We don't know $\pi_{j,0}, \pi_{j,1}, \ldots, \pi_{j,m+r}$ but we have estimated them, so ...
   - Compute the predicted values of X_{ji}, $\hat{X}_{ji}$, where $\hat{X}_{ji} = \hat{\pi}_{j,0} + \hat{\pi}_{j,1}Z_{1i} + \cdots + \hat{\pi}_{j,m}Z_{mi} + \hat{\pi}_{j,m+1}W_{1i} + \cdots + \hat{\pi}_{j,m+r}W_{ri}, i = 1, \ldots, n.$

Why W's are included? We want to test whether instruments can predict X's after controlling W's

Two Stage Least Squares (2SLS)

2. Replace X_{ji} by $\hat{X}_{ji}$ in the regression of interest: regress Y on $\hat{X}_{1i}, \ldots, \hat{X}_{ki}$ and $W_{1i}, \ldots, W_{ri}$ using OLS:

$$Y_i = \beta_0 + \beta_1 \hat{X}_{1i} + \cdots + \beta_k \hat{X}_{ki} + \beta_{k+1} W_{1i} + \cdots + \beta_{k+r} W_{ri} + u_i \quad (4)$$

   - Because $\hat{X}_{ji}$ is uncorrelated with u_i in large samples, so the first least squares assumption holds
   - Thus, $\beta_0, \beta_1, \ldots, \beta_{k+r}$ can be estimated by OLS using regression (4)
   - This argument relies on large samples (so $\pi_{j,0}, \pi_{j,1}, \ldots, \pi_{j,m+r}$ are well estimated using regression (3))
   - This the resulting estimator is called the "Two Stage Least Squares" (2SLS) estimator. We do not call it as IV estimator unless $k = m$

Estimation Results: Institution and Income

TABLE 4—IV REGRESSIONS OF LOG GDP PER CAPITA

	Base sample (1)	Base sample (2)	Base sample without Neo-Europes (3)	Base sample without Neo-Europes (4)	Base sample without Africa (5)	Base sample without Africa (6)	Base sample with continent dummies (7)	Base sample with continent dummies (8)	Base sample, dependent variable is log output per worker (9)
Panel A: Two-Stage Least Squares									
Average protection against expropriation risk 1985–1995	0.94	1.00	1.28	1.21	0.58	0.58	0.98	1.10	0.98
	(0.16)	(0.22)	(0.36)	(0.35)	(0.10)	(0.12)	(0.30)	(0.46)	(0.17)
Latitude		−0.65		0.94		0.04		−1.20	
		(1.34)		(1.46)		(0.84)		(1.8)	
Asia dummy							−0.92	−1.10	
							(0.40)	(0.52)	
Africa dummy							−0.46	−0.44	
							(0.36)	(0.42)	
"Other" continent dummy							−0.94	−0.99	
							(0.85)	(1.0)	

V. How to test whether an IV is good or not?

Check Instrument Validity

Checking instrument relevance. The model with a single included endogenous regressor:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 W_{1i} + \cdots + \beta_{1+r} W_{ri} + u_i \, .$$

The first stage regression:

$$X_i = \pi_0 + \pi_1 Z_{1i} + \cdots + \pi_m Z_{mi} + \pi_{m+1} W_{1i} + \cdots + \pi_{m+r} W_{ri} + v_i \, .$$

- The IVs are relevant if one of $\pi_1, \ldots, \pi_m$ is nonzero. Sometimes, we say the instruments are strong
- If $\pi_1, \ldots, \pi_m$ are all zero, the IVs are said to be irrelevant
- The IVs are said to be weak if all $\pi_1, \ldots, \pi_m$ are nearly zero (how close?)

Check Instrument Validity

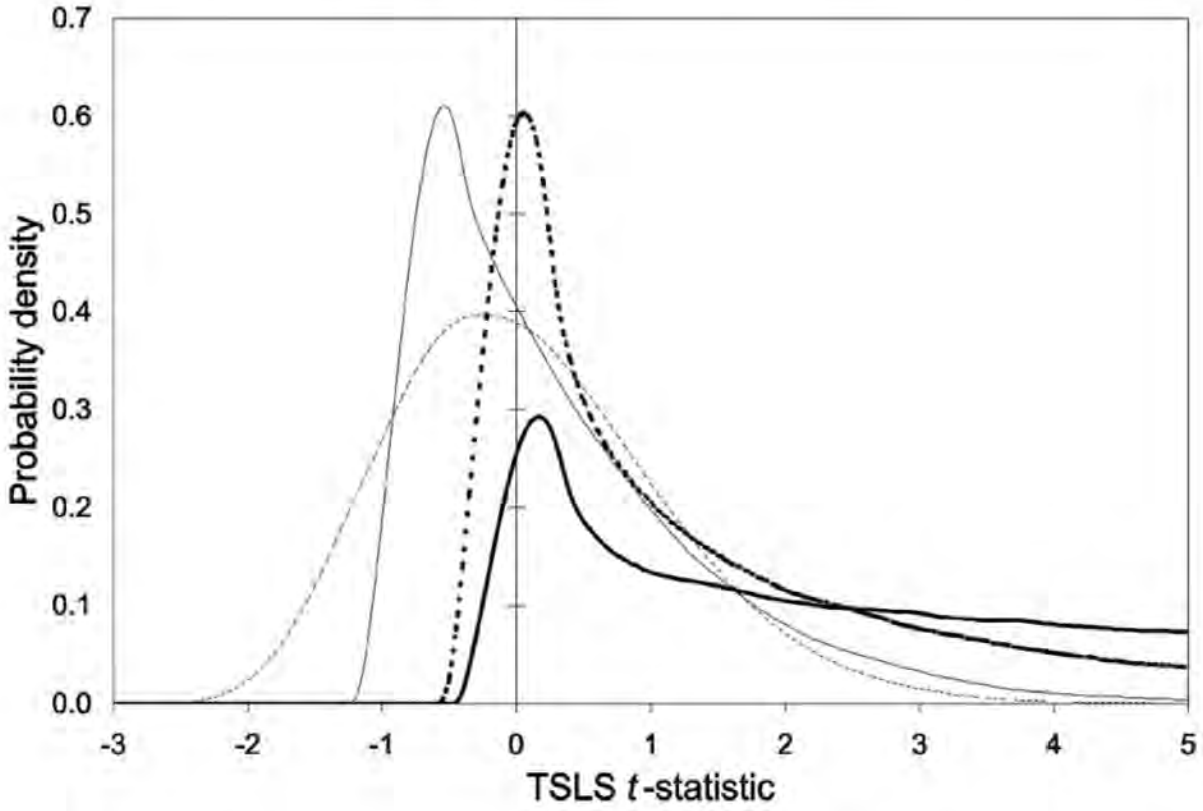
How to test for relevance condition? It amounts to testing for

$$H_0 : \pi_1 = \ldots = \pi_m = 0 .$$

- It can be done by F test in the first stage regression
- When $m = 1$, we reject H_0 if $F > 3.84$ at 5% level. We may conclude that the relevance condition is likely to be true
- However, this does not imply that the sampling distribution of the 2SLS/IV estimator is normal for large n
- In applications, people usually require that $F > 10$ for $m = 1$. If $F < 10$, then the IV is weak. This is the definition of weak IV
- The definition of weak IVs for multiple instruments is given by Stock and Yogo (2005)

Check Instrument Validity

An example: the sampling distribution of $\hat{\beta}_{1,IV}$ with weak IV



Dark line: irrelevant IV; dashed light line: strong IV

Relevance: First Stage

	Base sample (1)	Base sample (2)	Base sample without Neo-Europes (3)	Base sample without Neo-Europes (4)	Base sample without Africa (5)	Base sample without Africa (6)	Base sample with continent dummies (7)	Base sample with continent dummies (8)	Base sample, dependent variable is log output per worker (9)
Panel B: First Stage for Average Protection Against Expropriation Risk in 1985–1995									
Log European settler mortality	−0.61	−0.51	−0.39	−0.39	−1.20	−1.10	−0.43	−0.34	−0.63
	(0.13)	(0.14)	(0.13)	(0.14)	(0.22)	(0.24)	(0.17)	(0.18)	(0.13)
Latitude		2.00		−0.11		0.99		2.00	
		(1.34)		(1.50)		(1.43)		(1.40)	
Asia dummy							0.33	0.47	
							(0.49)	(0.50)	
Africa dummy							−0.27	−0.26	
							(0.41)	(0.41)	
"Other" continent dummy							1.24	1.1	
							(0.84)	(0.84)	
R^2	0.27	0.30	0.13	0.13	0.47	0.47	0.30	0.33	0.28

Check Instrument Validity

Checking instrument exogeneity. Recall that all the instruments must be uncorrelated with the error term. That is, $Cov(Z_{1i}, u_i) = 0, \ldots, Cov(Z_{mi}, u_i) = 0$.

Consider the simplest case

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$
$$X_i = \pi_0 + \pi_1 Z_i + v_i \, .$$

The natural option is to check whether the sample covariance between Z_i and $\hat{u}_i = Y_i - \hat{\beta}_{0,IV} - \hat{\beta}_{1,IV} X_i$ is far away from zero.

- What is the problem with this method? What is $S_{Z\hat{u}} = \widehat{Cov}(Z, \hat{u}) = n^{-1} \sum_{i=1}^{n} (Z_i - \bar{Z})(\hat{u}_i - \bar{\hat{u}})$?

Check Instrument Validity

Testing for instrument exogeneity can be done in the case of overidentification. Consider the simple case

$$Y_i = \beta_0 + \beta_1 X_i + u_i \, .$$

Suppose there are two valid instruments: Z_{1i}, Z_{2i}.

- Then you could compute two separate 2SLS estimates
- Intuitively, if these two 2SLS estimates are very different from each other, then something must be wrong: one or the other (or both) of the instruments must be invalid.
- The J-test of overidentifying restrictions makes this comparison in a statistically precise way.
- This can only be done if $m > k$ (overidentified). The J-test is also known as overidentification test

Check Instrument Validity

The J-test is the Anderson-Rubin test, using the 2SLS estimator instead of the hypothesized value $\beta_{1,0}$. The recipe:

- Obtain the 2SLS estimator using all m instruments; compute the predicted values $\hat{Y}_i$ using the actual X's
- Compute the residual $\hat{u}_i = Y - \hat{Y}_i$
- Regress $\hat{u}_i$ on $W_{1i}, \ldots, W_{ri}, Z_{1i}, \ldots, Z_{mi}$
- Compute the F-statistic testing the hypothesis that the coefficients on $Z_{1i}, \ldots, Z_{mi}$ are all zero
- The J-statistic is $J = mF$

Check Instrument Validity

Distribution of the J-statistic

- Under the null hypothesis that all the instruments are exogeneous, J has a chi-squared distribution with $m-k$ degrees of freedom
- If $m = k$, $J = 0$ (does this make sense?)
- If some instruments are exogenous and others are endogenous, the J statistic will be large, and the null hypothesis that all the instruments are exogeneous will be rejected

Exogeneity

Key Assumption: settler mortality in the past has no direct effect on current economic performance

Test:

- Directly control for many of the variables that could plausibly be correlated with both settler mortality and economic outcomes, and checking whether the addition of these variables affects our estimate
- Overidentification test (caution: such tests may not lead to a rejection if all instruments are invalid, but still highly correlated with each other)

Adding additional controls

TABLE 5—IV REGRESSIONS OF LOG GDP PER CAPITA WITH ADDITIONAL CONTROLS

	Base sample (1)	Base sample (2)	British colonies only (3)	British colonies only (4)	Base sample (5)	Base sample (6)	Base sample (7)	Base sample (8)	Base sample (9)
Panel A: Two-Stage Least Squares									
Average protection against expropriation risk, 1985–1995	1.10 (0.22)	1.16 (0.34)	1.07 (0.24)	1.00 (0.22)	1.10 (0.19)	1.20 (0.29)	0.92 (0.15)	1.00 (0.25)	1.10 (0.29)
Latitude		−0.75 (1.70)				−1.10 (1.56)		−0.94 (1.50)	−1.70 (1.6)
British colonial dummy	−0.78 (0.35)	−0.80 (0.39)							
French colonial dummy	−0.12 (0.35)	−0.06 (0.42)							0.02 (0.69)
French legal origin dummy					0.89 (0.32)	0.96 (0.39)			0.51 (0.69)
p-value for religion variables							[0.001]	[0.004]	[0.42]

Adding additional controls

TABLE 6—ROBUSTNESS CHECKS FOR IV REGRESSIONS OF LOG GDP PER CAPITA

	Base sample (1)	Base sample (2)	Base sample (3)	Base sample (4)	Base sample (5)	Base sample (6)	Base sample (7)	Base sample (8)	Base sample (9)
Panel A: Two-Stage Least Squares									
Average protection against expropriation risk, 1985–1995	0.84 (0.19)	0.83 (0.21)	0.96 (0.28)	0.99 (0.30)	1.10 (0.33)	1.30 (0.51)	0.74 (0.13)	0.79 (0.17)	0.71 (0.20)
Latitude		0.07 (1.60)		−0.67 (1.30)		−1.30 (2.30)		−0.89 (1.00)	−2.5 (1.60)
p-value for temperature variables	[0.96]	[0.97]							[0.77]
p-value for humidity variables	[0.54]	[0.54]							[0.62]
Percent of European descent in 1975			−0.08 (0.82)	0.03 (0.84)					0.3 (0.7)
p-value for soil quality					[0.79]	[0.85]			[0.46]
p-value for natural resources					[0.82]	[0.87]			[0.82]
Dummy for being landlocked					0.64 (0.63)	0.79 (0.83)			0.75 (0.47)
Ethnolinguistic fragmentation							−1.00 (0.32)	−1.10 (0.34)	−1.60 (0.47)

Adding additional controls

TABLE 7—GEOGRAPHY AND HEALTH VARIABLES

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
	Instrumenting only for average protection against expropriation risk						Instrumenting for all right-hand-side variables			Yellow fever instrument for average protection against expropriation risk	
	Panel A: Two-Stage Least Squares										
Average protection against expropriation risk, 1985–1995	0.69 (0.25)	0.72 (0.30)	0.63 (0.28)	0.68 (0.34)	0.55 (0.24)	0.56 (0.31)	0.69 (0.26)	0.74 (0.24)	0.68 (0.23)	0.91 (0.24)	0.90 (0.32)
Latitude		−0.57 (1.04)		−0.53 (0.97)		−0.1 (0.95)					
Malaria in 1994	−0.57 (0.47)	−0.60 (0.47)					−0.62 (0.68)				
Life expectancy			0.03 (0.02)	0.03 (0.02)				0.02 (0.02)			
Infant mortality					−0.01 (0.005)	−0.01 (0.006)			−0.01 (0.01)		

Overidentification Test

TABLE 8—OVERIDENTIFICATION TESTS

	Base sample (1)	Base sample (2)	Base sample (3)	Base sample (4)	Base sample (5)	Base sample (6)	Base sample (7)	Base sample (8)	Base sample (9)	Base sample (10)
Panel A: Two-Stage Least Squares										
Average protection against expropriation risk, 1985–1995	0.87	0.92	0.71	0.68	0.72	0.69	0.60	0.61	0.55	0.56
	(0.14)	(0.20)	(0.15)	(0.20)	(0.14)	(0.19)	(0.14)	(0.17)	(0.12)	(0.14)
Latitude		−0.47		−0.34		0.31		−0.41		−0.16
		(1.20)		(1.10)		(1.05)		(0.92)		(0.81)
Panel B: First Stage for Average Protection Against Expropriation Risk										
European settlements in 1900	3.20	2.90								
	(0.62)	(0.83)								
Constraint on executive in 1900			0.32	0.26						
			(0.08)	(0.09)						
Democracy in 1900					0.24	0.20				
					(0.06)	(0.07)				
Constraint on executive in first year of independence							0.25	0.22		
							(0.08)	(0.08)		
Democracy in first year of independence									0.19	0.17
									(0.05)	(0.05)
R^2	0.30	0.30	0.20	0.24	0.24	0.26	0.19	0.25	0.26	0.30
Panel C: Results from Overidentification Test										
p-value (from chi-squared test)	[0.67]	[0.96]	[0.09]	[0.20]	[0.11]	[0.28]	[0.67]	[0.79]	[0.22]	[0.26]